

Waste minimization and pollution prevention in university laboratories

By Robin M. Izzo

You do it every time you sign a manifest. By signing the manifest, you are certifying that you have a program in place to minimize generation of hazardous waste at your institution. It's right there in section 16. But how can you do this for research laboratories?

Coordinating hazardous waste management at a university can be very complex due to a number of factors, including:

- Most colleges and universities are decentralized, such that most academic and administrative departments function independent of each other.
- Most research laboratories have an unstable workforce, relying heavily on graduate students and postdoctoral associates who are usually at the university for a limited time.
- The type of operations and waste generation changes frequently as the focus of the research changes.
- For many universities, there is no central purchasing or chemical stockroom.

Hazardous waste generation at Princeton University has continuously decreased over the last five years due to a number of programs initiated by

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both the office of Environmental Health and Safety (EHS) and the specific academic and administrative departments themselves. Disposal costs have decreased by more than 60% over that time. During our last inspection by the New Jersey Department of Environmental Protection in August 1999, our program was highly commended for its success in reducing waste by more than 50% at one of our sites.

Although our success is due to a combination of a number of efforts, the common thread is to foster partnerships between EHS staff and waste generators and to keep the procedures as simple as possible. Because even the best ideas cannot work effectively if the information does not reach the waste generator, much of the focus of the hazardous waste program is educating and informing the generators about safe and efficient ways to manage their waste. This is primarily accomplished by an extensive web page for waste disposal; training of all faculty, staff, and students; an e-mail discussion list; and a waste newsletter.

THE WASTE DISPOSAL PROGRAM

Princeton University is comprised of approximately 4600 undergraduates, 1750 graduate students, 1000 faculty, and 4000 administrative and support staff. There are approximately 500 research laboratories in 12 departments. Other regular waste generators include the maintenance department, the cogeneration plant, construction, and visual arts, which includes theater prop shops and set design.

The laboratory waste disposal program is coordinated through the university industrial hygienist with assistance from other EHS staff members and an environmental compliance manager. Unlike many institutions of a similar size, there is not a full-time hazardous waste manager, nor is there a central waste collection facility. Indeed, the combined time of all individuals involved in managing the waste disposal program is less than one full-time equivalent.

Generally, waste pickups are scheduled the last Thursday of every month. Waste generators are required to transport their wastes to a pickup point. Two waste contractors collect, package, and ship the waste to offsite treatment, storage, and disposal facilities. Although there is one 90-day storage area on campus in the chemistry department, there is rarely any waste stored in the room except on the day of the pickup.

Having two waste companies service the university at the same time allows us to minimize the time needed for waste pickups; allows more choices for disposal based on method or cost, or both; and minimizes the need to leave a waste on campus due to a contractor that cannot transport a particular waste on that day.

The EHS staff overseeing the pickup will review all waste containers to look for disposal trends, unusual chemicals or unusual quantities, and opportunities for bulking similar wastes for cost savings. The university industrial hygienist also reviews all paperwork to look for trends and collect information that might be useful

to target areas that need help with minimization. While collection of this type of information might be useful before the pickup, for example, by having the generator notify us of the wastes beforehand or using a two-part tag system, this method has proven to be useful and saves a great deal of time. We do encourage generators to call our office for advice on disposal of unusual wastes.

EHS staff also collects information from waste generators through annual laboratory surveys and through their questions. All of the information helps us to work with waste generators to find ways to minimize waste.

Cost Management

Waste disposal costs are paid through central funds managed by EHS. Individual departments or laboratories are not charged for regular disposal, in order to encourage them to use the system. However, the department or laboratory is charged for disposal or other fees under certain circumstances:

- The department or laboratory would be responsible for paying fines resulting from notices of violations imposed by regulators for their improper waste management in their department or laboratory. Because the laboratory workers are well trained and have easy access to many resources for proper waste disposal, there is no excuse for poor management.
- Any costs attributed to poor waste management in a laboratory are charged back to the particular laboratory or department. One example of this might be beyond their expiration date ethers that require special disposal.
- The cost of characterization of unknowns is charged back to the department. The department may, in turn, collect payment from the individual investigator.
- The laboratory or department pays labor costs when contractors are needed to clean out laboratories, which sometimes occurs when a researcher is getting ready to leave or when a laboratory group moves from one facility to another.

GETTING THE WORD OUT

Even the best waste management idea will not be completely successful unless everyone knows about it and understands it. In a decentralized environment like a university, getting the word out can be challenging and a single method of communication is unlikely to be effective. At Princeton University, several methods are used to communicate to waste generators. If an individual does not give attention or is not reached by one method, he or she will likely be reached by one of the others.

The Waste-Paper

The *Waste-Paper*, pictured in Figure 1, is a newsletter for waste generators. It is generally no more than two pages long and serves two purposes. The first is to notify waste generators of the exact date and any special details for the upcoming waste pickup. The second is to give waste generators information that can help them manage their wastes more efficiently and/or more safely, give waste minimization tips, and describe programs underway at Princeton.

The *Waste-Paper* has four regular columns: a featured article, "Waste Minimization Tips," "Joan's Pet Peeves" (what went right and what went wrong during the last waste pickup—Joan Hutzly is the technical specialist who helps with the waste pickups and bulks solvents), and the "Question of the Month." Not every issue contains all four columns, but the columns help us write and organize each issue.

Each issue is sent out two weeks before the waste pickup and is distributed in several different ways, depending on departmental preference. Some prefer sending a hard copy to each individual or to the contact person for distribution, whereas others prefer receiving their copies via e-mail. The current issue and all back issues are also available on our web page at <http://www.princeton.edu/~ehs/wastepaper/TOC.html>.

The newsletter has been very successful. Web statistics show that more people will access our waste disposal web pages in the days following distribution of the newsletter. We also tend to receive more phone calls looking

for waste management assistance after the newsletter goes out. The added bonus is that the New Jersey Department of Environmental Protection considers the newsletter to be a form of refresher training for waste generators.

In all, each issue normally takes approximately four hours to write, edit, and distribute. Finding information to include in each issue is usually not difficult. There are a multitude of excellent articles in health and safety periodicals and newsletters, including *Chemical Health & Safety*.

EHS Web Page

EHS has an extensive web page filled with practical information for safety and waste management. In particular, the "Laboratory Safety Manual" and the "Waste Disposal" section are excellent resources for laboratory workers. The waste disposal pages contain the most requested information, including general waste disposal procedures, a list of non-hazardous wastes, recycling information, and disposal information for some specific materials, such as ethidium bromide, oils, and mercury.

E-mail Listserve

EHS has an internal health and safety electronic mailing list (listserve) called SAFETYLINE. All of our contacts in each department are subscribed to the list. As with all listserves, any e-mail sent to the listserve, including replies, is distributed to all subscribers. The listserve is particularly useful for giving updates, sharing time-sensitive information, and collecting responses. It also allows for discussion and debate of issues.

Education and Training

All faculty and staff using chemicals are required to attend either hazard communication training or laboratory safety training. Both include instruction for safe handling of chemicals, cleaning up spills, and managing hazardous waste.

Incoming science and engineering graduate students are required to attend laboratory safety training during their first month at Princeton, usually just after registration. These courses are given on a department-by-department

The Waste-Paper

The Hazardous Waste Disposal Monthly Update

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Combined Pick-Up for November-December

Because of historically light pickups in November and December, EHS will combine the waste pickup for these months into one pickup on Thursday, December 16. Please make sure to bring all your accumulated waste to this month's pickup (10/28), rather than wait over a month until next time.

Question of the Month – How Do I Clean Up a Mercury Spill Properly?

Begin by picking up the droplets. Use an index card or scraper to consolidate the droplets, and pick up the pool using a pipette, syringe or vacuum pump. Do not use the house vacuum system without a charcoal filter trap. Small droplets can be picked up with adhesive tape or wet paper towels.

Commercial products such as sponges and powders may also be used. The sponges are typically not very effective. Sulfur is not a very effective means for cleaning up mercury.

Place the mercury in a glass or plastic jar or a sturdy plastic bag. Only add *visibly* contaminated debris. Seal the bag and affix a label identifying the material as "mercury spill debris".

Please make sure to minimize the amount of debris involved. If gloves or other debris do not visibly contain mercury, they do not need to be included with the other waste.

Inform Joan Hutzly of EHS at 8-6251 or hutzly@princeton.edu about the waste. Joan will ensure it is collected for recycling.

If you have any questions contact EHS at 8-5294 for assistance.

Joan's Pet Peeves



Well, you did it again. You made Joan mad. She really tries to stay happy on waste day, but there are some things that just set her off:

- *Formulas on waste stickers* – what Joan kindly refers to as "alphabet soup". When filling out your waste stickers, you must list the chemical constituents with their IUPAC names. For example, Hydrochloric Acid (good)...HCl (bad).
- *Debris in Waste Containers* – You don't want to be there when Joan finds this. Solid debris such as gloves, paper towels, etc. should be thrown in the regular trash (if not grossly contaminated) or placed in a bag and sealed. Most liquid wastes are poured into larger containers, and debris in the bottle makes this task even more difficult.

New Spill Kits



kstates@princeton.edu

Some of you may have noticed some new additions in E-Quad and Bowen Hall. EHS has placed spill kits on the 100 level outside of each elevator. The kits include a large blue pail of vermiculite and a 5-gallon pail containing spill control materials (gloves, aprons, goggles, plastic liners, dustpan & brush, and spill cleanup instructions).

Similar spill kits will soon be placed in the Frick-Hoyt Chemistry complex.

EHS will provide these kits for a *central location only* at no charge to the department. Please contact Kelly States at 8-2648 or for more information.

This month's waste pickup will be held on Thursday
October 28, 1999

Figure 1. The Waste-Paper.

ment basis, allowing us to give examples and information most appropriate and useful for the particular group.

Undergraduates declare their major just before their junior year. Juniors are required to take a "core lab" course in their major. EHS provides the same training given to faculty, staff, and graduate students to these undergraduates as part of the curriculum, usually during their first laboratory meeting. Because these students will be doing some independent laboratory work for their jun-

ior and senior thesis, it is important that they receive training to be able to work in a situation where their work is not constantly supervised.

Laboratory Surveys and Visits

For the past two years, technicians conducting the annual fume hood surveys perform a limited laboratory inspection concurrently and visit laboratories without fume hoods just for the inspection. The technicians received training on how to conduct the

survey and work from checklists. Although the focus of the survey changes a bit each year, laboratory waste disposal practices are always included. Survey results for all laboratories in a department are sent to the "Department Safety Manager" (our department contact) and e-mailed to each individual principal investigator. As a direct result, there has been a significant improvement in labeling containers, keeping them sealed, and other related activities.

Nothing works better than having direct conversations with waste generators right at their point of generation—the laboratory. Our measure of success for not just waste management, but all health and safety issues, is the number of minutes it takes us to walk through a building. Because getting stopped by several people who recognize us and who have questions lengthens the trip, the longer it takes, the more we feel that the word is getting out.

To strengthen this, we try to visit one waste-generating building each week to take a look around, pop in on people, and search for potential waste accumulation problems. Receiving room and support staff tend to be founts of information. We also invite interested individuals to quarterly safety manager breakfasts, giving everyone an opportunity to talk about waste and other safety issues in an informal setting. This follows the “if you feed them, they will come” philosophy.

SUCCESSFUL ENDEAVORS

Defining “Waste”

Not every chemical waste needs to be disposed as hazardous waste. Many of the inquiries that we receive are from workers who want to know whether a particular chemical must be treated as hazardous waste. Stanford University put together a great list of non-hazardous wastes and how to properly dispose of them. This list is available on their web site at <http://www.stanford.edu/dept/EHS/waste/nonhaz.html>.

Seeing this as a great benchmarking opportunity, we created a very similar list for our web page, after checking with our local sewer treatment authorities and our environmental compliance manager. Laboratory workers and our waste disposal contractor have been very cooperative in packaging these materials into boxes and disposing them as regular trash or drain disposing, as appropriate. We also encourage laboratory workers to minimize the amount of personal protective equipment disposed as hazardous waste, limiting it to grossly contaminated equipment and those used with highly hazardous materials.

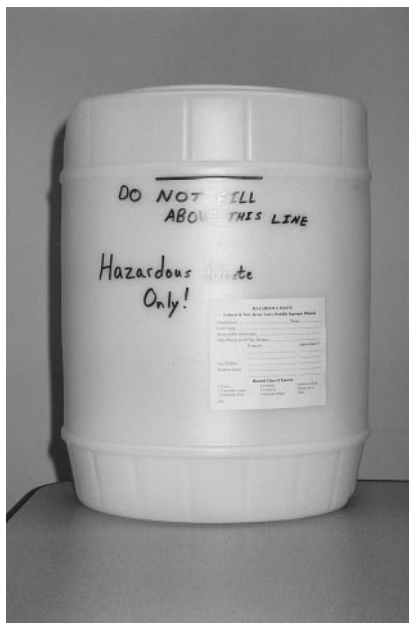


Figure 2. Five-Gallon Containers Used for Waste Collection.

Bulking into Drums

EHS identified several waste streams that could easily be bulked into drums rather than laboratory packing. These include photographic chemicals, solvents, vacuum pump oil, and contaminated silica gel. EHS supplies five-gallon carboys, such as those in Figure 2, for waste collection and encourages their use. The day before the waste pickup, the laboratory workers bring the filled containers down to the waste collection area, where an EHS technician transfers the waste into 55-gallon drums. The emptied five-gallon containers are made available to the laboratories again. Bulking is performed in diked, well-ventilated areas with grounding strips available and takes no more than three hours, once each month.

This approach resulted in almost 50% cost savings and more than 30% reduction of the weight of the waste, compared to labpacking the same materials. To further the use of this system, we will visit laboratories that bring down chemicals in large glass containers and personally bring them a five-gallon carboy to try for their next waste collection.

Scaling Down Experiments

Based on what we see during the waste pickup, EHS targets certain laboratories for a review of their procedures. The review focuses on ways to scale down experiments, consider an end-of-procedure step to make the product non-hazardous waste, or substitute with a less hazardous material. Usually, just opening such a discussion helps the researcher find better ways, because the researcher was probably not thinking in terms of disposal when designing the experiment.

Minimizing Waste from Teaching Laboratories

All teaching laboratories use student manuals. EHS reviews all manuals annually to recommend ways to scale down experiments, substitute with less toxic materials, and ensure that the procedures include instruction on safety precautions and waste handling. Many courses have switched to microscale techniques for most of their experiments, and a few have moved to computer modeling. The biology and microbiology courses have been able to completely eliminate the use of acrylamide and ethidium bromide and generate very little chemical or medical waste.

Minimizing Mercury Waste

Several years ago, EHS led an effort to replace as many mercury thermometers as possible with non-mercury thermometers, preferably with coated glass. As a result, the amount of mercury and debris from broken thermometers has decreased in those areas. At the time, not all thermometers could be replaced, due to limitations in the response rate and temperature scale allowed by non-mercury thermometers. This is less of an issue with the advent of digital thermometers. Now that New Jersey considers mercury a Universal Waste, we are able to collect mercury wastes from all of our EPA ID areas and consolidate it to minimize the amount of debris. Both efforts resulted in a reduction of the amount of mercury waste and the cost of recycling. We also identified researchers who are willing to reuse some of the collected mercury.

Table 1. Waste Stream Chart

Category	Description	Container	Disposition
Nonrecyclable waste glass	Plate glass, light bulbs, broken glass, chemical containers, noncontaminated serological pipettes	Tall cardboard container with plastic liner	Solid waste
Empty chemical containers	Intact, triple-rinsed glass and plastic (#1 or 2), no caps	Plastic bucket with half lid	Recyclable
Lab Trash	Noncontaminated gloves, bench paper, packaging materials, foil, plastic bags, paper towels, weighing boats, media, culture plates, centrifuge tubes, filter flasks	Standard waste can	Solid waste
Sharps	Pasteur and other glass pipettes, needles, syringes, scalpel blades, razor blades, slides, coverslips	Sharps container	Medical waste
Medical waste	Experimentally cultured stocks, plates, or other materials meeting New Jersey definition. Gels containing >0.1% ethidium bromide (dark pink or red)	Cardboard medical waste box with red plastic liner	Medical waste
Nonhazardous chemical waste	Dry chemical waste included on nonhazardous list (see web page).	Sealed cardboard box labeled, "Nonhazardous Waste"	Solid waste
Chemical waste	All other chemical wastes, per EPA definitions.	Original container or compatible container, sealed and labeled with yellow waste sticker	Chemical waste

Laboratory Checkouts

Many of our science and engineering departments formalized their checkout procedure for outgoing students, postdocs, faculty, and staff, some threatening to hold a diploma or paycheck if the procedure is not followed. EHS also offers to have our waste contractors clean out the chemicals from the laboratory, as long as the department pays the labor cost. As a result, the number of unknowns and unneeded chemicals left in the laboratory has been significantly reduced.

Snooping in the Trash

During the annual laboratory surveys, our technicians were asked to look at the "trash" in each laboratory: the regular trash, the medical waste boxes, and the chemical waste. We found that most chemical containers were disposed of as regular solid waste, many non-hazardous chemicals were being disposed as hazardous waste, and many items that are not considered medical or medical-like waste were going into the medical waste boxes.

To promote recycling and reduce the amount of chemical and medical waste, laboratory wastes were classified into seven categories: non-recyclable waste glass, empty chemical containers, sharps, medical waste, hazardous chemical waste, non-hazardous chemical waste, and laboratory trash. A more detailed description may be found in Table 1.

Non-recyclable glass (e.g., Pyrex and broken glass) is collected into cardboard boxes that are sealed by laboratory workers and collected by the janitors for disposal in the solid waste stream. Empty chemical containers are rinsed as directed and placed in glass recycling bins. Sharps are collected in sharps containers and disposed as medical waste. Medical waste is collected in red-bag lined cardboard boxes for disposal through a medical waste contractor. Hazardous chemical wastes are disposed through the hazardous waste system. Non-hazardous chemical wastes, as listed on our web page, are labeled as non-hazardous and placed in regular

laboratory trash or drain disposed, as appropriate.

These efforts realized a reduction of medical waste tonnage by half and a cost savings of more than \$10,000, while still promoting pollution prevention. The laboratory workers are more aware of what and how they dispose of, as well.

THE FUTURE

EHS continues to look for ways to reduce the amount of hazardous waste in an environmentally sound manner. We constantly update our web page and other information sources to offer useful information to laboratory workers and strive to make more and more information available utilizing electronic means. We are considering a chemical exchange program and plan an on-line waste tutorial. Most importantly, we continue to foster partnerships with the departments and laboratories to prevent pollution, minimize waste production, and enhance safety in the laboratory.